COMPOSITION OF RELATIVE CYCLES IS A RELATIVE CYCLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base-change compatibility of the composition of families of cycles (relative-cycles chapter, Lemma lemma-construction-base-change, canonical tag 0H6K). cited at proof lines 3009-3010. Reduces the specialization check to the case S the spectrum of a discrete valuation ring.
- Locality of the composition on opens and compatibility of specialization maps with flat pullback (relative-cycles chapter, Lemma lemma-construction-opens, canonical tag 0H6J, and Lemma lemma-specialization-flat-pullback, canonical tag 0H4L). cited at proof lines 3014-3018. The constructions commute with restriction, so X and Y may be assumed affine.
- Proper completely decomposed cover putting a relative cycle into the image of the cycle-classes map (relative-cycles chapter, Lemma lemma-get-cycles, canonical tag 0H59) and the cycle-classes construction (relative-cycles chapter, Equation equation-cycle-classes, canonical tag 0H58). cited at proof lines 3020-3023. Provides the completely decomposed proper morphism g : Y' to Y with g^*alpha in the image.
- Envelope decomposition of a prime cycle along a completely decomposed morphism (Chow Homology, Lemma chow-lemma-envelope, canonical tag 0GU8). cited at proof lines 3026-3029. Writes beta_eta as a sum of push-forwards g_{eta,*}beta'_eta along the fibres of the cover.
- Functoriality of relative cycles under pushforward and the uniqueness of extensions agreeing on a dense open (relative-cycles chapter, Lemma lemma-relative-cycle-functoriality, canonical tag 0H51, and Lemma lemma-uniqueness-extension, canonical tag 0H55). cited at proof lines 3032-3035 and 3044-3045. The pieces beta' assemble to a relative cycle whose pushforward equals beta, and relative cycles are detected by their generic fibres; functoriality also reduces the target to the pullback composition.
- Bilinearity of the composition and its push-pull formula (relative-cycles chapter, Lemma lemma-construction-bilinear, canonical tag 0H6I, and Lemma lemma-construction-push-pull, canonical tag 0H6N). cited at proof lines

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3036-3037 and 3042-3043. Linearity reduces to a single summand; push-pull moves the pushforward f_* outside the composition.
- Discrete valuation ring version of the cover lemma (relative-cycles chapter, Lemma lemma-get-cycles-dvr, canonical tag 0H5A). cited at proof lines 3046-3047. Further reduces the pulled-back composition to a single cycle class of a flat closed subscheme.
- Closed-pair formula for the composition and the specialization lemma for families associated to closed subschemes (relative-cycles chapter, Lemma lemma-construction-closed, canonical tag 0H6M, and Lemma lemma-family-associated-closed-specialization, canonical tag 0H57). cited at proof lines 3054-3060. Computes alpha composition beta as the cycle class of the flat fibred product and certifies it is a relative cycle.

Source reference. Original lemma 0H6K: sources/stacks-chow/tags.txt, source lines 21425–21425. Cited at proof lines 3009-3010: reduces to the discrete valuation ring case.

Source reference. Original lemma 0H6J: sources/stacks-chow/tags.txt, source lines 21424–21424. Cited at proof lines 3015-3016: lets the proof work locally on Y and X.

Source reference. Original lemma 0H4L: sources/stacks-chow/tags.txt, source lines 21356–21356. Cited at proof lines 3016-3018: the specialization construction commutes with restriction.

Source reference. Original lemma 0H59: sources/stacks-chow/tags.txt, source lines 21379–21379. Cited at proof lines 3020-3023: provides the cover g with g^*alpha in the image.

Source reference. Original equation 0H58: sources/stacks-chow/tags.txt, source lines 21378–21378. Cited at proof lines 3022-3023: the image condition on g^*alpha.

Source reference. Original lemma 0GU8: sources/stacks-chow/tags.txt, source lines 20993–20993. Cited at proof lines 3026-3029: decomposes beta_eta into pushforwards along the fibres of the cover.

Source reference. Original lemma 0H51: sources/stacks-chow/tags.txt, source lines 21371–21371. Cited at proof lines 3032-3033 and 3044-3045: g_*beta' and beta are relative cycles, and the target reduces to the pullback composition.

Source reference. Original lemma 0H55: sources/stacks-chow/tags.txt, source lines 21375–21375. Cited at proof line 3035: g_*beta' equals beta because they agree at eta.

Source reference. Original lemma 0H6I: sources/stacks-chow/tags.txt, source lines 21423–21423. Cited at proof lines 3036-3037: reduces to a single summand of the envelope decomposition.

Source reference. Original lemma 0H6N: sources/stacks-chow/tags.txt, source lines 21428–21428. Cited at proof lines 3042-3043: moves the pushforward f_* outside the composition.

Source reference. Original lemma 0H5A: sources/stacks-chow/tags.txt, source lines 21380–21380. Cited at proof lines 3046-3047: reduces to a single flat cycle class.

Source reference. Original lemma 0H6M: sources/stacks-chow/tags.txt, source lines 21427–21427. Cited at proof lines 3054-3057: computes the composition explicitly.

Source reference. Original lemma 0H57: sources/stacks-chow/tags.txt, source lines 21377–21377. Cited at proof lines 3059-3060: certifies the computed class is a relative cycle.

Lemma 0.2. *If α and β are relative cycles, then so is $\alpha \circ \beta$.*

Let S be a locally Noetherian scheme. Let $X \to Y$ be a morphism of schemes locally of finite type over S. We are going to define a map

$$z(X/Y, r) \otimes_{\mathbf{Z}} z(Y/S, e) \longrightarrow z(X/S, r + e), \quad \alpha \otimes \beta \longmapsto \alpha \circ \beta$$

Proof. The formation of $\alpha \circ \beta$ is compatible with base change by Lemma 0H6K. Thus we may assume S is the spectrum of a discrete valuation ring with generic point η and closed point 0 and we have to show that $sp_{X/S}((\alpha \circ \beta)_\eta) = (\alpha \circ \beta)_0$. Since we are trying to prove an equality of cycles, we may work locally on Y and X (this uses Lemmas 0H6J and 0H4L to see that the constructions commute with restriction). Thus we may assume X and Y are affine. By Lemma 0H59 we can find a completely decomposed proper morphism $g : Y' \to Y$ such that $g^*\alpha$ is in the image of (0H58).

Since the family of morphisms $g_\eta : Y'_\eta \to Y_\eta$ is completely decomposed, we can find $\beta'_\eta \in Z_e(Y'_\eta)$ such that $\beta_\eta = \sum g_{\eta,*}\beta'_\eta$, see Chow Homology, Lemma 0GU8. Set $\beta'_0 = sp_{Y'/S}(\beta'_\eta)$ so that $\beta' = (\beta'_\eta, \beta'_0)$ is a relative e-cycle on Y'/S. Then $g_*\beta'$ and β are relative e-cycles on Y/S (Lemma 0H51) which have the same value at η and hence are equal (Lemma 0H55). By linearity (Lemma 0H6I) it suffices to show that $\alpha \circ g_*\beta'$ is a relative $(r + e)$-cycle.

Set $X' = X \times_Y Y'$ and denote $f : X' \to X$ the projection. By Lemma 0H6N we see that $\alpha \circ g_*\beta' = f_*(g^*\alpha \circ \beta')$. By Lemma 0H51 it suffices to show that $g^*\alpha \circ \beta'$ is a relative $(r + e)$-cycle. Using Lemma 0H5A and bilinearity this reduces us to the case discussed in the next paragraph.

Assume $\alpha = [Z/X/Y]_r$ and $\beta = [W/Y/S]$ where $Z \subset X$ is a closed subscheme flat and of relative dimension $\leq r$ over Y and $W \subset Y$ is a closed subscheme flat and of relative dimension $\leq e$ over S. By Lemma 0H6M we see that

$$\alpha \circ \beta = [Z \times_X W/X/S]_{r+e}$$

and $Z \times_X W \subset X$ is a closed subscheme flat over S of relative dimension $\leq r + e$. This is a relative $(r + e)$-cycle by Lemma 0H57. □